Mathematics practice booklet: Grade 1 expectations

Grade 1 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 1 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: 1.OA.A.1, 1.OA.A.2, 1.OA.B.3, 1.OA.B.4, 1.OA.C.5, 1.OA.C.6, 1.OA.D.7, 1.OA.D.8

A worked example

Maryam has 8 dates. Her brother gives her 5 more dates. How many dates does she have now?

1. Start with 8 dates.
2. Count on 5 more: 9, 10, 11, 12, 13.
3. 8 + 5 = 13.

13 dates

Questions

1. What is 7 + 5?

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2. Hamdan has AED 9. He finds AED 4 more. How many dirhams does he have?

   A. 12

   B. 13

   C. 14

3. What is 15 - 6?

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4. There are 14 date cookies on a plate. Layla and her sister eat 5 of them. How many are left?

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5. Find the missing number: 8 + __ = 15

   A. 7

   B. 8

   C. 23

6. Salem had some dirhams. He spent AED 4 and has AED 6 left. How many dirhams did he have at first?

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7. Add three numbers: 3 + 4 + 6. Hint: look for two numbers that make 10.

   A. 10

   B. 12

   C. 13

8. Is 6 + 4 = 4 + 6 true or false?

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9. Which fact is in the same family as 6 + 7 = 13?

   A. 13 - 6 = 7

   B. 13 + 6 = 19

   C. 7 - 6 = 1

10. You know 8 + 8 = 16. Use it to find 8 + 9.

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11. Which is bigger, 9 + 4 or 7 + 5?

    A. 7 + 5

    B. They are equal

    C. 9 + 4

Section 2: Shapes, measuring and space

Standards practised: 1.G.A.1, 1.G.A.2, 1.G.A.3, 1.MD.A.1, 1.MD.A.2, 1.MD.B.3

A worked example

A round flatbread is cut into 4 equal pieces. What is each piece called? How many pieces make the whole bread?

1. Four equal pieces means each piece is one of four equal shares.
2. Each piece is called a quarter.
3. All 4 quarters together make the whole bread.

Each piece is a quarter; 4 quarters make the whole

Questions

1. How many sides does a triangle have?

   A. 2

   B. 3

   C. 4

2. How many corners does a square have?

   _____

3. Maha holds an AED 1 coin. It is flat and round. What shape is it?

   _____

4. A door has 4 sides: two long sides and two short sides. What shape is it?

   A. A square

   B. A triangle

   C. A rectangle

5. Maha puts two square tiles side by side so they touch. What bigger shape can she make?

   _____

6. Noura cuts a sandwich into 2 equal pieces. What is each piece called?

   _____

7. A pizza is cut into 4 equal slices. What is each slice called?

   A. A quarter

   B. A half

   C. The whole

8. Salem's pencil is 9 cubes long. Maryam's pencil is 6 cubes long. Which pencil is longer, and by how many cubes?

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9. A ribbon is 5 paper clips long. A string is 8 paper clips long. Which one is shorter?

   A. The string

   B. The ribbon

   C. They are the same length

10. On a clock the short hand points to 3 and the long hand points to 12. What time is it?

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11. On a clock the short hand is between 4 and 5 and the long hand points to 6. What time is it?

    A. 4:06

    B. 6:04

    C. 4:30

12. Is a half of a bread bigger than a quarter of the same bread? Say why.

Section 3: Data, graphs and chance

Standards practised: 1.MD.C.4

A worked example

Noura counted the fruit in a bowl: 4 apples, 6 oranges and 3 bananas. How many more oranges than apples are there?

1. Oranges: 6. Apples: 4.
2. Take away: 6 - 4 = 2.

2 more oranges

Questions

1. Children chose a favourite fruit: apples 5, bananas 3, oranges 4. How many children chose in all?

2. In the same count, which fruit did the most children choose?

   A. Bananas

   B. Apples

   C. Oranges

3. Which fruit did the fewest children choose?

4. How many more children chose apples than bananas?

   A. 2

   B. 3

   C. 8

5. How many fewer children chose bananas than oranges?

6. How many children chose oranges or bananas?

7. Hind counted shapes in a picture: 4 circles, 7 squares, 3 triangles. How many squares are there?

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8. How many shapes are there in all?

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9. How many more squares than triangles are there?

   A. 10

   B. 3

   C. 4

10. Which two kinds of shape together make exactly 10?

    A. Squares and triangles

    B. Circles and squares

    C. Circles and triangles

Section 4: Numbers and calculating

Standards practised: 1.NBT.A.1, 1.NBT.B.2, 1.NBT.B.3, 1.NBT.C.4, 1.NBT.C.5, 1.NBT.C.6

A worked example

How many tens and how many ones are in the number 36?

1. The first digit, 3, is the tens: 3 tens is 30.
2. The second digit, 6, is the ones: 6 ones.
3. 30 + 6 = 36.

3 tens and 6 ones

Questions

1. Count on: 118, 119, __. What number comes next?

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2. How many tens and ones are in 52?

   A. 2 tens and 5 ones

   B. 5 tens and 2 ones

   C. 52 tens

3. What number is 6 tens and 3 ones?

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4. Which number is bigger, 48 or 84? Say it with the words bigger or smaller.

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5. Omar says 29 is bigger than 31 because 9 is bigger than 1. Is he right? Explain with tens.

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6. What is 10 more than 34?

   A. 24

   B. 35

   C. 44

7. Layla has 58 stickers and gives 10 of them to her cousin. How many stickers does she have left?

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8. Add: 43 + 5

   A. 48

   B. 93

   C. 47

9. Salem has AED 35. His grandmother gives him AED 20 more. How many dirhams does he have now?

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10. Khalid has AED 70 and spends AED 30. How many dirhams does he have left?

    A. 4

    B. 40

    C. 100

11. Maryam has four AED 10 notes. How many dirhams is that?

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. Count on 5 from 7: 8, 9, 10, 11, 12. So 7 + 5 = 12.
2. B. 13. Count on 4 from 9: 10, 11, 12, 13. If your child chose A (12), they counted the 9 as one of the 4 steps.
3. 15 - 5 = 10, then 10 - 1 = 9. So 15 - 6 = 9.
4. 14 - 4 = 10, then 10 - 1 = 9. So 9 cookies are left.
5. A. 7. 8 + 7 = 15, because 15 - 8 = 7. If your child chose C (23), they added 8 and 15 instead of finding the missing part.
6. He spent 4 and has 6 left, so put them together: 4 + 6 = 10. He had AED 10.
7. C. 13. 4 + 6 makes 10, then 10 + 3 = 13. If your child chose A (10), they added only the 4 and the 6.
8. 6 + 4 and 4 + 6 both make 10, so it is true. Adding in any order gives the same total.
9. A. 13 - 6 = 7. The numbers 6, 7 and 13 make a family, and 13 - 6 = 7 uses the same three. If your child chose B (13 + 6 = 19), they added the total instead of taking away.
10. 8 + 9 is one more than 8 + 8. 16 and 1 more is 17.
11. C. 9 + 4. = 13 and 7 + 5 = 12, and 13 is bigger. If your child chose B (equal), they did not work out both totals.

Section 2: Shapes, measuring and space

1. B. 3. Count the sides of a triangle: 1, 2, 3. If your child chose C (4), they may have thought of a square.
2. Touch each corner of a square and count: 1, 2, 3, 4. So a square has 4 corners.
3. A coin is flat and round, so its shape is a circle. A circle has no corners.
4. C. A rectangle. Two long sides and two short sides make a rectangle. If your child chose A (a square), they missed that a square has 4 equal sides.
5. Two squares side by side make one long shape with 4 corners, which is a rectangle.
6. Cutting into 2 equal pieces makes halves. So each piece is a half.
7. A. A quarter. 4 equal slices means each slice is one quarter. If your child chose B (a half), they used the name for 2 equal slices.
8. Subtract to compare: 9 - 6 = 3. So Salem's pencil is longer by 3 cubes.
9. B. The ribbon. 5 is less than 8, so the ribbon with 5 clips is shorter. If your child chose A (the string), they picked the longer one.
10. The short hand on 3 gives the hour and the long hand on 12 means o'clock. So it is 3 o'clock.
11. C. 4:30. The long hand on 6 means half past, and the short hand is past 4, so it is 4:30. If your child chose A (4:06), they read the 6 as minutes.
12. A half is the same as 2 quarters, and 2 quarters is more than 1 quarter. So a half is bigger.

Section 3: Data, graphs and chance

1. Add the three groups: 5 + 3 + 4 = 12. So 12 children chose.
2. B. Apples. have 5, which is bigger than 4 and 3, so most children chose apples. If your child chose C (oranges), they picked the second biggest number.

3. Look for the smallest number: 3 is less than 4 and 5. So bananas, with 3.
4. A. 2. Take away to compare: 5 - 3 = 2. If your child chose C (8), they added 5 and 3 instead.
5. Take away to compare: 4 - 3 = 1. So 1 fewer child chose bananas.
6. Add the oranges and the bananas: 4 + 3 = 7 children.
7. Look at the squares in the count. The number for squares is 7.
8. Add all three groups: 4 + 7 = 11, then 11 + 3 = 14.
9. C. 4. Take away to compare: 7 - 3 = 4. If your child chose A (10), they added 7 and 3 instead.
10. A. Squares and triangles. 7 + 3 = 10, so it is squares and triangles. If your child chose B (circles and squares), they added 4 and 7 to get 11.

Section 4: Numbers and calculating

1. Count on one more after 119. After 119 comes 120.
2. B. 5 tens and 2 ones. In 52 the first digit is the tens, so 5 tens, and the second digit is the ones, so 2 ones. If your child chose A (2 tens and 5 ones), they swapped the digits.
3. 6 tens is 60, and 3 ones is 3. So 60 + 3 = 63.
4. Compare the tens first: 84 has 8 tens and 48 has 4 tens. So 84 is bigger and 48 is smaller.
5. Compare the tens first: 31 has 3 tens and 29 has 2 tens. So 31 is bigger, and Omar is not right.
6. C. 44. 10 more is 1 more ten, so 3 tens becomes 4 tens and the answer is 44. If your child chose B (35), they added 1 and not 10.
7. Take 1 ten away: 5 tens becomes 4 tens. So 58 - 10 = 48 stickers.
8. A. 48. Add the ones: 3 + 5 = 8, so 43 + 5 = 48. If your child chose B (93), they added the 5 to the tens.
9. Add the tens: 3 tens + 2 tens = 5 tens, which is 50. Then 50 + 5 = 55.
10. B. 40. 7 tens - 3 tens = 4 tens, which is 40. If your child chose C (100), they added 70 and 30.
11. Count by tens four times: 10, 20, 30, 40. So she has 40 dirhams.